

Artur lasenovets

Information Security and Applied Cryptography Engineer

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PROFILE	Information security engineer and researcher focused on applied cryptography and privacy-enhancing technologies. I develop in Go , Python , and TypeScript ; I have basic experience with C++ and Rust . I conduct formal security analysis of cryptographic protocols and develop systems for private data search, retrieval, and processing based on homomorphic encryption .	
EDUCATION	Chongqing University of Posts and Telecommunications , Chongqing, China <i>School of Cyberspace Security and Information Law</i> ◇ M.S. in Network and Information Security	Sep. 2024–Jul. 2027 (expected)
	◇ Supervisor: Prof. Tang Fei ◇ GPA: 3.98/4.00	
	Ukhta State Technical University , Ukhta, Russia ◇ B.S. in Information Systems and Technologies ◇ Weighted GPA: 4.49/5.00	Mar. 2022–Jun. 2024
PUBLICATIONS	Published Works and Preprints Artur lasenovets , Fei Tang , H. Zhu , P. Wang , and L. Liu . “ Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval ,” arXiv:2511.02656, 2025. Manuscript under review. H. Zhu , F. Tang , W. Qin , J. Shan , P. Wang , and Artur lasenovets . “ EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption ,” <i>Journal of King Saud University – Computer and Information Sciences</i> , 2025. Artur lasenovets and Fei Tang . “ MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework ,” <i>International Research Journal</i> , no. 1 (163), 2026. Piyumi M. S. Rajapaksha* , Artur lasenovets* , Yinxue Yi , and Fei Tang . “ TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection ,” 2026. Manuscript under review. *Equal contribution.	
AWARDS AND ACHIEVEMENTS	Chinese Government Scholarship (CSC) for Graduate Studies	2024–2027
	Full scholarship for the entire M.S. programme at CQUPT.	
	Winner, EdTech Project Competition	2023
	Won the competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base.	
	First Degree Diploma, Severgeoecontech	2022
	Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference.	
	Sber Student Accelerator Regional Demo Day	2023
	Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western District regional demo day.	
	Supporting certificates	

EXPERIENCE SKILLS	Graduate Student Researcher @ Applied Cryptography Group, CQUP Private Information Retrieval for Hyperledger Fabric - Designed a private-read mechanism for the Hyperledger Fabric ledger world state based on a PIR protocol using the BGV HE scheme from the Lattigo library. - Developed a threat model with an honest-but-curious (HBC) node and demonstrated query privacy based on the IND-CPA security of BGV, including repeated queries and observable information leakage. Privacy-Preserving Aggregation for Federated Reinforcement Learning - Designed the TWAPPA-FRL architecture and implemented a private weighted aggregation mechanism for DQN updates based on the CKKS HE scheme using the TenSEAL library. - Developed a threat model with an honest-but-curious (HBC) aggregator and demonstrated client-update privacy based on the IND-CPA security of CKKS, including observable information leakage. Verifiable Multi-Keyword Searchable Encryption - Analyzed the threat model with an HBC data owner and user and a malicious cloud server; demonstrated query privacy in the real/ideal model based on OT security and the pseudorandomness of CI-rOPRF/PRG under explicitly defined search- and access-pattern leakage. - Analyzed the justification of result verifiability using a file-key verification tag and AEAD, which detect key and file tampering by a malicious cloud server. Security Intern (EDR, TI, SOC) @ Positive Technologies - Developed Python tools with unit-test coverage for validating Windows paths, network configurations, and endpoint-control policies. - Built an Airflow-based malware-processing pipeline: collected samples from MalwareBazaar, stored artifacts in S3, and scanned them with YARA. - Analyzed EVTX logs, PowerShell, and malicious VBA macros; extracted IoCs and mapped TTPs to MITRE ATT&CK for CVE-2020-5902 and CVE-2023-38831. Software Engineer @ AcademAI - Implemented a Python REST API and a LangChain-based course-generation pipeline using asynchronous LLM orchestration. - Developed the application layer with Next.js, TypeScript, Prisma, and Auth.js, including authentication and persistent storage of course data. - Configured CI/CD using Docker and GitHub Actions. Standards and regulations: GOST R 34.10/34.11, GOST 34.12/34.13; Federal Laws Nos. 63, 149, 152, 187; FIPS 203–205 Privacy-enhancing technologies: Private Information Retrieval (PIR/CPiR), Searchable Symmetric Encryption (SSE) Homomorphic encryption: Brakerski–Gentry–Vaikuntanathan (BGV), Brakerski–Fan–Vercauteren (BFV), Cheon–Kim–Kim–Song (CKKS); IND-CPA, IND-CCA2, and EUF-CMA security analysis Post-quantum cryptography: ML-KEM, ML-DSA, SLH-DSA; LWE/RLWE, Module-LWE, SIS/Module-SIS	Sep. 2024–Present Aug. 2024–Feb. 2025 Aug. 2023–Aug. 2024 Libraries and frameworks: Lattigo, Microsoft SEAL, OpenFHE, TenSEAL, feanor, fheanor Systems and infrastructure: Linux, Proxmox VE, VMware, Docker, GitHub Actions, GitLab CI/CD, Nginx, Ansible, Airflow, S3, Hyperledger Fabric Security engineering: YARA, Chainsaw, malware and log analysis, IoC extraction, TTP mapping to MITRE ATT&CK Languages: Russian – native; English – C1; Chinese – HSK 3
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